

## **Pneumothorax**

Collection of air located within the thoracic cage between visceral and parietal pleura.

Two main types:

### **1) Spontaneous Pneumothorax**

- **Primary:** In the absence of underlying lung disease
- **Secondary:** With underlying lung disease (e.g. CF, ILD, necrotizing pneumonia)

### **2) Traumatic Pneumothorax**

- Due to blunt, crush, or penetrating trauma to the chest, injury from a diagnostic or therapeutic procedure, or mechanical ventilation

## **History and Physical Examination**

- Acute onset pleuritic chest pain
- Acute onset shortness of breath
- Vital signs: Tachycardia, tachypnoea, oxygen desaturation
- Look out for signs of tension pneumothorax: Severe respiratory distress, tachycardia, impaired peripheral perfusion, hypotension, raised jugular venous pressure (JVP), pulsus paradoxus
- Look out for signs of mediastinal shift: Tracheal deviation, displaced apex beat
- Look out for related or associated features e.g. Marfanoid features, clubbing, Harrison's sulci
- Examination findings: Ipsilateral reduced chest expansion, hyperresonant percussion note, reduced air entry, reduced vocal resonance. There may be subcutaneous emphysema

## **Investigations**

- Chest X-ray (CXR)
  - Pleural line, hyperlucency and attenuation of vascular and lung markings on the affected side
  - Ipsilateral lung collapse;  $\pm$  flattening of ipsilateral hemidiaphragm
  - Look for mediastinal and tracheal shift to the contralateral side (away from the pneumothorax)
  - Look for associated pneumomediastinum or subcutaneous emphysema or (very rarely) pneumopericardium
  - Estimate size of pneumothorax – based on latest evidence, the size of the pneumothorax in itself is not a determinant for the need for chest drain insertion
  - ***Note that tension pneumothorax is a clinical diagnosis.***

## **Guidelines in Management**

### **1) *Tension pneumothorax***

- Immediate decompression (needle thoracostomy) followed by definitive drainage
- Tension pneumothorax is a clinical diagnosis. Do not wait for imaging if clinically suspected. There is no need for a blood gas before decompression of a tension pneumothorax.

### **2) *Secondary pneumothorax***

- Patients with secondary pneumothorax should be admitted for management and observation with referral to the respiratory service.

### **3) *Primary spontaneous pneumothorax***

- Conservative management of patients with a mostly asymptomatic primary spontaneous pneumothorax (even large pneumothoraces) is non-inferior to interventional management with a chest drain<sup>1</sup>.
- Indications for chest drain insertion would include:
  - Tension pneumothorax
  - Haemodynamic instability (e.g. hypotension)
  - Hypoxia or cyanosis
  - Respiratory distress
  - Significant symptoms e.g. pain despite adequate analgesia, worsening breathlessness

### **4) *Recurrent Pneumothorax***

- In patients with recurrence of a pneumothorax after initial resolution of a prior pneumothorax, referral to thoracic surgeons for surgical intervention is recommended. This, by itself, is NOT an indication for a chest tube insertion unless the above indications are present.

### **5) *Note on the Use of Oxygen Supplementation via Non-Rebreather Mask (NRM)***

- Supplemental oxygen may be administered via a non-rebreather mask to promote theoretical nitrogen washout and facilitate/enhance pleural air resorption, although evidence supporting its clinical benefit is limited.

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<sup>1</sup> NEJM 2020 Conservative versus Interventional Treatment for Spontaneous Pneumothorax: 98.5% in the intervention group (chest tube inserted) had resolution within 8 weeks, compared to 94.4% in the conservatively managed group.

#### 6) **Discharge Advice**

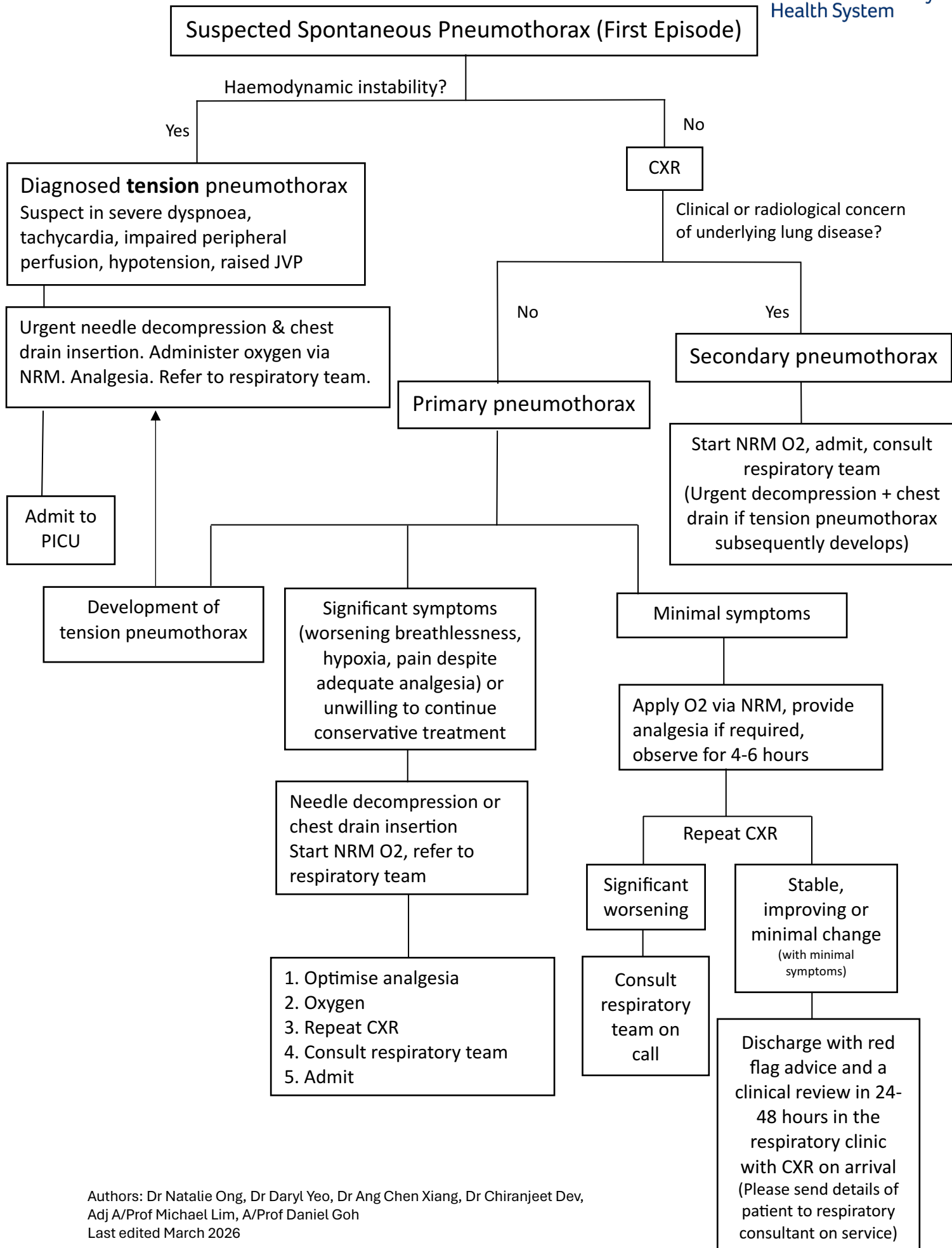
- Return to the emergency department if there is recurrence or worsening of symptoms, including severe chest pain (despite adequate analgesia), shortness of breath and increased work of breathing. Although progression to tension pneumothorax is uncommon (estimated ~1-2%)<sup>2</sup> in primary spontaneous pneumothorax, patients should still seek urgent medical attention if symptoms worsen.
- No scuba diving, bungee jumping or parachuting. If participation is desired, consider specialist referral for counselling and possible risk-reducing procedures.
- No airline travel for at least 1 week after radiological resolution<sup>3</sup>
- No contact sports (e.g. football, rugby, wrestling, hockey) for 4-6 weeks<sup>4</sup>
- No heavy exercise at sea level for 4-6 weeks e.g. running, swimming, weightlifting<sup>4</sup>
- No exercise at altitude for 4-6 weeks e.g. skiing, mountain climbing<sup>4</sup>
- No smoking or vaping (active and/or passive)

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<sup>2</sup> Kelly C, Carlberg M, Madsen T. Spontaneous pneumothorax resulting in tension physiology. Am J Emerg Med. 2019: Several papers estimate 1-2% of spontaneous pneumothorax will be under tension.

<sup>3</sup> BTS Clinical Statement on Air Travel for Passengers with Respiratory Disease (2022); Joint ERS/EACTS/ESTS Clinical Practice Guidelines on Adults with Spontaneous Pneumothorax (2024)

<sup>4</sup> Based on local practice, avoidance of contact sports/heavy exercise for 4-6 weeks is advised. There is currently no official recommendation on the specific duration to avoid contact sports/heavy exercise.



References:

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2. Lieu N, Ngo P, Chennapragada SM, et al. Update in management of paediatric primary spontaneous pneumothorax. *Paediatr Respir Rev*. 2022;41:73-79.  
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3. Roberts ME, Rahman NM, Maskell NA, et al. British Thoracic Society Guideline for Pleural Disease Thorax 2023;78:1143-1156.
4. Walker S, Hallifax R, Ricciardi S, Fitzgerald D, Keijzers M, Lauk O, et al. Joint ERS/EACTS/ESTS clinical practice guidelines on adults with spontaneous pneumothorax. *European Respiratory Journal*. 2024;63(5):2300797. doi:10.1183/13993003.00797-2023.
5. Kelly C, Carlberg M, Madsen T. Spontaneous pneumothorax resulting in tension physiology. *Am J Emerg Med*. 2019;37(1):173.e1–173.e2.  
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6. Coker RK, Armstrong A, Church AC, Holmes S, Naylor J, Pike K, et al. BTS Clinical Statement on air travel for passengers with respiratory disease. *Thorax*. 2022;77(4):329-350. doi:10.1136/thoraxjnl-2021-218110.